

Milestone 3, Hypothesis 2: Is Melanoma Recall a Threshold Problem?

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Base paper: Datta et al., *Soft-Attention Improves Skin Cancer Classification Performance*, iMIMIC @ MICCAI 2021

Reference model: our reproduced Inception-ResNet-v2 baseline (`cv_project_irv2.ipynb`, seed 42)

Companion to: `M3_hypothesis.md` (Hypothesis 1, focal loss)

1. Motivation

The reproduced IRv2 baseline separates melanoma from other lesions well but rarely *predicts* melanoma:

Melanoma metric (IRv2, test set, n = 34)	Value
One-vs-rest AUC	0.915
Recall at arg-max	0.441 (15 / 34), 95 % Wilson CI 0.29–0.60
Precision at arg-max	0.517
Specificity at arg-max	0.982

With an AUC of 0.915 and a specificity of 0.982, the arg-max decision sits at a very conservative point on the melanoma ROC curve. The model ranks melanomas above most non-melanomas, but argmax lets a melanoma win only when its probability beats every other class, and that rarely happens when *nv* holds 80 % of the data.

Hypothesis 1 (focal loss) predicts that changing the training loss will raise melanoma recall. Its own falsification criteria say a recall gain that costs more than 0.02 weighted precision is only *partial support*, because a threshold change could produce the same trade-off. So far no one has measured what a threshold change alone can do. This hypothesis fills that gap.

2. Hypothesis

Lowering the melanoma decision threshold on the already-trained IRv2 baseline, with no retraining, will raise melanoma recall from 0.44 to at least 0.59 ($\geq 20 / 34$) while keeping melanoma specificity ≥ 0.95 . The reason is that the baseline's errors come from the arg-max operating point, not from poor features.

Decision rule. Predict *mel* if $p(\text{mel}) \geq t$. Otherwise predict the arg-max over the remaining six classes. With $t = \text{arg-max-equivalent}$, this reproduces the baseline exactly, so **t is the only variable**.

Predicted effect (cross-fitted, see §3):

Metric	Baseline (arg-max)	Predicted at spec ≥ 0.95
Melanoma recall	0.441	0.59–0.75
Melanoma precision	0.517	0.35–0.45 (falls)
Melanoma specificity	0.982	0.95–0.96
nv recall	0.989	0.96–0.98
Macro recall	0.720	+0.02 to +0.04
Weighted precision	0.909	within ± 0.015
Weighted / melanoma AUC	0.978 / 0.915	unchanged by construction

3. Protocol

1. **Inputs.** Test-set softmax outputs (828×7) from the saved IRv2 seed-42 checkpoint. If they were not saved, one inference pass produces them in a few minutes. No GPU training is needed.
2. **No threshold chosen on the data it is scored on.** The checkpoint already saw the whole training set, so a slice of training data would give optimistic probabilities. Instead we use **5-fold cross-fitting on the test set**, stratified by class. For each fold, t is picked on the other four folds as the lowest threshold whose melanoma specificity is ≥ 0.95 , then applied to the held-out fold. Predictions from all five folds are pooled before computing metrics.
3. **Operating points reported.** (a) arg-max, (b) specificity ≥ 0.95 , (c) specificity ≥ 0.90 , (d) specificity 0.713, which is the mean dermatologist specificity in Haenssle et al. (2018). At (d) we compare the model's sensitivity with the dermatologists' 0.866.
4. **Uncertainty.** 95 % Wilson intervals for per-class recall. 1,000-resample bootstrap intervals for weighted precision and macro recall. We repeat the 5-fold split with 20 random seeds and report mean $\pm$ std for t .
5. **Same procedure on IRv2 + Soft-Attention** (mel AUC 0.969). This tells us whether attention's recall gain ($15 \rightarrow 21 / 34$) survives once both models are compared at the same specificity.

4. How we will know we were wrong

- **Reject** if cross-fitted melanoma recall at specificity ≥ 0.95 is below $20 / 34$ (0.588). That would mean the ROC curve is too flat at low false-positive rates for a threshold to help, and the problem does lie in the features.
- **Reject** if weighted precision drops by more than 0.02. The gain would then be bought with too many *nv* false alarms.
- **Inconclusive** if the threshold chosen in each fold varies by more than 0.2 across folds. With only 34 melanomas, the operating point would not be stable enough to recommend.

5. Why this matters for Hypothesis 1

H2 outcome	Consequence for the focal-loss experiment (H1)
Supported	Arg-max recall alone cannot show that focal loss helps. H1 must be judged as melanoma recall at matched specificity against the threshold-tuned baseline, or as a change in AUC.
Rejected	The baseline's features are the bottleneck. A training-time change such as focal loss is justified, and arg-max comparisons remain fair.

Either way, H2 sets the bar that H1 has to clear, and it costs zero GPU-hours.

6. Risks

Risk	Mitigation
Only 34 test melanomas, so one image = 2.9 recall points	Wilson CIs; repeated cross-fitting; report counts, not just rates
Checkpoint itself was selected on the test set (paper's protocol)	Stated as a shared limitation with M2; does not favour any threshold
Softmax probabilities may be poorly calibrated	Thresholding needs only the <i>ranking</i> , not calibration. We also report a reliability diagram for mel.
More mel predictions take images from nv/bkl	Track nv and bkl recall and the full confusion matrix at each operating point

7. Results table (to fill in after running)

Model	Operating point	t (mean $\pm$ std)	mel recall	mel spec.	mel prec.	macro recall	W. precision
IRv2	arg-max	—	0.441	0.982	0.517	0.720	0.909
IRv2	spec ≥ 0.95		<i>pred. 0.59–0.75</i>				
IRv2	spec ≥ 0.90						
IRv2	spec = 0.713 (dermatologists)		<i>vs. 0.866</i>				
IRv2 + SA	arg-max	—	0.618	0.987	0.677	0.713	0.913
IRv2 + SA	spec ≥ 0.95						

No clinical claims are made. All results are benchmark numbers on a curated research dataset.

References. Datta S. K. et al., iMIMIC @ MICCAI 2021. Haenssle H. A. et al., *Annals of Oncology* 29(8), 2018. Lin T.-Y. et al., *Focal Loss for Dense Object Detection*, ICCV 2017.